

TriVoKo

Phase 4 - Checkout & payments

Time	Start date	Finish date	Status
2 weeks			Not started

Goal: One cart, one Stripe payment, one order split into one package per seller - with stock held safely, coupons, emails and a PDF invoice.

Real-life example: At a food court you pay once at the main counter, and your token is sent to the dosa stall and the juice stall separately. Each stall only sees its own part of your order.

Step checklist

★ = you type the key logic (Claude explains first, then reviews). "You" = only you can do it.

#	Step	Who	Done
1	Flyway: orders, packages, order_items (copy title, variant and price), payments, processed_payment_events, coupons, coupon_redemptions	Claude	☐
2	Money helpers (rupees ↔ paise, rounding) with unit tests	Claude	☐
3	Coupon rules: flat or %, minimum order, expiry, one use per customer; admin create endpoint	Claude	☐
4	★ You type the ORDER SPLIT : group cart lines by seller into packages, shipping fee per package (free above ₹499, else ₹40), coupon share per item (by price, rounding remainder on the last item)	★ Madhavan	☐
5	★ You write at least one test for the order split (2 sellers + coupon → exact amounts)	★ Madhavan	☐
6	CheckoutService: recheck products, sellers, coupon and prices; hold stock with UPDATE ... SET stock = stock - :qty WHERE stock >= :qty; roll back and name the item that ran out	Claude	☐
7	Copy the EventHub PaymentGateway switch (Fake when no Stripe key, Stripe Checkout when set)	Claude	☐
8	Stripe webhook (signature check) + "verify on return" fallback + idempotency → order PAID, packages PLACED, coupon used, cart emptied	Claude	☐
9	Expiry job: unpaid after 10 minutes → EXPIRED, stock back; late payment re-takes stock or is refunded automatically	Claude	☐
10	notify module (copy from EventHub): bell + emails after commit with retry; order email to customer, new-package email to each seller	Claude	☐
11	PDF invoice with OpenPDF: GET /api/orders/{number}/invoice.pdf	Claude	☐
12	Frontend: checkout page (address, coupon, summary per package, "price changed" notice), payment result, My orders, order detail	Claude	☐
13	Concurrency test: 100 threads try to buy 10 units → exactly 10 orders	Claude	☐
14	Create the Stripe test account, put sk_test in .env, install the Stripe CLI	You	☐

#	Step	Who	Done
15	Real Stripe test: pay with 4242 4242 4242 4242 and receive a real signed webhook (the EventHub to-do)	Together	☐

Output of this phase

- Working checkout with Stripe test mode
- Split orders with packages per seller
- Order emails in Mailpit and PDF invoices
- Your own OrderSplitter class + test

Done when (check before moving on)

Check	Passed
Pay once for items from 2 sellers → 2 packages PLACED	☐
The 100-thread stock test passes	☐
A real Stripe 4242 payment confirmed by a signed webhook	☐

Explain it back (in your own words)

Question	Done
Walk through checkout from "Place order" to "PLACED" - which table changes at each step?	☐
How is the coupon shared between packages, and why store it on each item?	☐
Why can a webhook arrive twice, and what stops a double order?	☐

Finish routine

Check	Done
All tests pass and CI is green	☐
Committed and pushed	☐
Done-when checks shown working in the browser	☐
Session summary + SESSION_LOG written	☐

Skills you practise: Transactions, atomic SQL updates, money maths, Stripe Checkout, webhooks, idempotency, scheduled jobs, PDF

Where your work goes: `backend\...\order`, `...\payment`, `...\coupon`, `...\notify`; checkout pages in `frontend`

My notes
